02 — Business Model Canvas

Value creation, revenue, cost, and market — at a glance

💡 **KEY TAKEAWAY**

Six B2B revenue lines anchored on a tiered manufacturer SaaS subscription, sold first into NRW's dense polymer/chemical cluster and expanded DACH/EU-wide as ESPR/PPWR compliance becomes mandatory everywhere.

VALUE PROPOSITIONS

- Verifiable — not self-declared — sustainability claims
- ESPR / PPWR / GS1 Digital Link compliance out of the box
- Industrial-grade, decimal-precise material data
- Cryptographic tamper-evidence
- One system for manufacturer → auditor → buyer trust

KEY PARTNERS

- RWTH Aachen / Fraunhofer research groups
- ISCC PLUS & REDcert2 certification bodies
- GS1 Germany (Digital Link standard)
- NRW chemical / plastics associations
- Neon (managed Postgres infrastructure)
- Independent auditors & auditing firms

KEY ACTIVITIES

- Platform engineering (schema, API, portal UX)
- Regulatory mapping as ESPR/PPWR evolve
- Auditor network onboarding & training
- Manufacturer onboarding & data migration

KEY RESOURCES

- The codebase (data model, ledger, QR engine)
- Embedded domain expertise (polymer science, EU compliance, security)
- Live production database & verified batch history
- Auditor & manufacturer pilot relationships

CUSTOMER RELATIONSHIPS

- Direct B2B sales with onboarding support
- Self-service supplier-invite links
- White-glove auditor onboarding
- Long-term account relationships

CHANNELS

- Direct sales into the NRW chemical corridor
- Industry association & cluster events
- Certification-body co-marketing
- The public QR-scan page as a brand touchpoint

CUSTOMER SEGMENTS

- **Primary:** mid-sized compounders & recyclers (50–500 employees), NRW/DACH
- **Secondary:** B2B buyers (OEMs, converters) needing verified material data
- **Tertiary:** independent auditors & certification-adjacent providers

COST STRUCTURE

- Engineering & product development (dominant, early-stage)
- Cloud infrastructure (Neon, hosting)
- Compliance & domain expertise (ongoing)
- Sales, partnerships & customer success
- Security/audit & certification maintenance
- Grant & compliance reporting overhead

REVENUE STREAMS

- ① Tiered manufacturer subscription (core)
- ② Auditor / verification seat licensing
- ③ B2B buyer / API access
- ④ Compliance export & certificate services
- ⑤ Onboarding, migration & integration services
- ⑥ Future: white-label / multi-tenant licensing

Revenue streams — the detail

The platform is designed around recurring, usage-aligned B2B SaaS revenue rather than one-off project fees, reflecting that a Digital Product Passport only has value if it stays live and current for the lifetime of every batch it describes.

- **Tiered manufacturer subscription** — the core revenue line, tiered by batch volume and feature access (base: passport creation, QR/GS1, public page; higher: genealogy, carbon calculator, ESG/PPWR analytics, green-tax estimation, bulk certificate/label generation).
- **Auditor/verification seat licensing** — auditors are a distinct, professionally liable user class, individually credentialed and priced separately.
- **B2B buyer/API access** — a stable, versioned Buyer API for OEMs and converters feeding verified supplier data into their own ERP/CSRD reporting, priced per connected account or data volume.
- **Compliance export & certificate services** — usage-based fees for PDF Certificates of Conformity, printable QR label sheets, and audit-ready compliance export workbooks with an embedded live hash-chain integrity check.
- **Onboarding, migration & integration services** — one-time professional-services fees for larger accounts, priced separately from recurring SaaS revenue.
- **(Future) White-label / multi-tenant licensing** — a plausible higher-value, lower-volume line once the multi-role architecture is proven at scale — flagged as a post-MVP option, not a Year 1 assumption.

Cost structure — the detail

Engineering and product development dominate the early-stage cost line, alongside cloud infrastructure (structurally low per-passport cost thanks to a stateless, compute-light architecture), ongoing compliance and domain expertise (a recurring cost, not a one-time setup, as ESPR/PPWR delegated acts continue to be published), sales and customer success (churn is existential here — a customer's entire batch history and public QR resolvers depend on continuity of service), periodic security/audit review, and the reporting overhead that comes with public co-financing.

Target market: NRW and EU

Immediate: NRW's polymer and chemical processing base — the Rhineland chemical corridor and Ruhr-area recycling sector — targeting compounders, recyclers, and converters (50–500 employees) already producing PCR/PIR or bio-based grades. **Adjacent:** because ESPR and PPWR are EU regulations, every polymer manufacturer across the EU faces the same deadline — NRW is the beachhead (home region, RWTH Aachen access, dense customer concentration), with a clear expansion path into DACH and then Benelux/broader EU.